

Shiqi (Stella) Chen

Computer Science and Information Security Specialist

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Profile

Software engineer with experience in backend systems, internal platform engineering, infrastructure, and CI/CD automation in a production healthcare environment. Strong academic background in computer science and information security, with interests in low-level systems, cryptography and machine learning.

Education

University of Toronto

Sep 2021 – Jun 2026

Honours Bachelor of Science with High Distinction

Toronto, Canada

- Computer Science Specialist
- Information Security Specialist
- Mathematical Science Minor
- CGPA: 3.95/4.00
- Dean's List Scholar (2022 Summer, 2023 Winter, 2026 Winter)

The Affiliated High School of South China Normal University (HFI)

Aug 2018 – Jun 2021

International Department, High School

Guangzhou, China

- Focused on mathematics, physics, and computer science.
- President of the Rubik's Club.

Sun Yat-sen University Affiliated Middle School

Sept 2015 – July 2018

Middle School

Guangzhou, China

Professional Experience

PocketHealth

May 2024 – Aug 2025

Software Engineering Intern (PEY)

Toronto, Canada

- Developed Prometheus-based monitoring and internal alerting to track backend service health, traffic patterns, and operational reliability across distributed systems.
- Automated monthly Azure storage utilization reporting with cron-based pipelines, improving infrastructure cost visibility and reducing manual tracking for the finance team.
- Implemented JWT authentication to secure service-to-service communication across a distributed backend architecture.
- Designed and optimized dynamic GitLab CI/CD pipelines, modularizing infrastructure workflows and reducing deployment overhead.
- Maintained Kubernetes-based backend infrastructure on Azure, supporting containerized services, ingress controllers, deployment automation, and internal platform operations.
- Improved backend security and reliability by addressing vulnerabilities identified through automated security scanning and infrastructure review processes.
- Collaborated with platform and backend engineers to maintain deployment pipelines, troubleshoot

infrastructure issues, and improve reliability of containerized services in QA environment.

Kua.ai

Jun 2023 – Aug 2023

Web Integration Intern

Zhuhai, China

- Built an integration that connected Midjourney capabilities into the company's web platform despite the absence of an official public API.
- Applied reverse-engineering techniques to understand Discord-based API behavior and translate it into an internal integration workflow.
- Collaborated with teammates to refine implementation details and improve reliability of the final user-facing experience.

Teaching Experience

University of Toronto

Jan 2023 – Present

Teaching Assistant

Toronto, Canada

- Served as a Teaching Assistant for undergraduate mathematics and computer science courses at the University of Toronto.
- Conducted tutorials, held weekly office hours, and provided instructional support to help students develop rigorous conceptual understanding and problem-solving ability.
- Supported course administration through invigilation and grading, with an emphasis on consistency, clarity, and fairness in assessment.
- Selected courses include MAT102, MAT135, MAT136, MAT223*, MAT224, MAT233, CSC236, CSC263, CSC347, and CSC427.

* Served as Head Teaching Assistant (TA Coordinator) for MAT223.

Selected Coursework and Academic Interests

Work Study

University of Toronto

- **Research Assistant** (Supervisor: Prof. Qun Wang) Jan 2025 – Apr 2025
 - Built a model for the N-Vortex simulator to support simulation-based analysis and experimentation; validated and refined the implementation through testing, debugging, and iterative improvement.
- **Lecture Notes Assistant** (Supervisor: Prof. Lisa Jeffrey) Jan 2024 – Apr 2024
 - Prepared and refined lecture notes in elementary Lie theory, and strengthened my background in Lie algebra through independent study of the underlying theorems and proofs.

Selected Coursework

University of Toronto

- Systems: CSC358 Principle of Computer Networks; CSC368 Computer Architecture; CSC369 Operating Systems.
- Security and Cryptography: CSC347 Information Security; CSC427 Computer Security; MAT302 Introduction to Algebraic Cryptography.
- AI and Machine Learning: CSC311 Introduction to Machine Learning; CSC384 Introduction to Artificial Intelligence; CSC413 Neural Networks and Deep Learning; CSC490 Capstone Design Course Machine Learning for Vision.

- Distributed Systems and Blockchain: CSC2125 Blockchain Technology and Engineering

Academic Interests

Current focus

- Distributed systems, computer security, cryptography, blockchain systems, group theory, abstract algebra.

Technical Skills

Programming	Go, Kotlin, Java, Python, C/C++, TypeScript, UNIX/Shell, HTML/CSS
Tools	Git, L ^A T _E X, Overleaf, R Markdown, Linux/UNIX workflows
Core Areas	Backend development, CI/CD automation, systems, security, technical documentation
Languages	English, Mandarin, Cantonese